

Abstract Mapping Guide

XR Score Viewer · RayNeo X3 Pro · current rayNeoPort version

This document explains what the current abstract particle-mesh visualisation is showing. The visual uses five mapped streams from the sensor selected in the HUD.

Overall Reading

- **Slow global shape** shows tonic electrodermal activity.
- **Particle colour and directional trails** show temperature rate of change.
- **Local sparks, rings, and short deformations** show skin conductance responses.
- **Global particle size, synchronized pulse, and four corner halos** show heart rate.
- **Spatial dispersion and mesh coherence** show engagement.

Each stream owns a separate visual channel so that one stream does not suppress another. In particular, engagement does not reduce the particles' minimum size or visibility, leaving heart-rate cues readable when engagement is low.

The four physiological inputs are signed standardized signals. The runtime maps the expected z-score range from -3 to $+3$ into the visual range from -1 to $+1$ and may amplify recent variation to keep small changes legible. Values near zero represent baseline. Positive and negative values indicate position relative to baseline; they do not necessarily mean that the raw measurement is rising or falling at that instant. Engagement is a separate value from 0 to 1.

Behavioural Signal

Engagement

Engagement controls how dispersed or coherent the mesh is.

- **Low engagement:** particles disperse and drift, and the surface loses spatial coherence.
- **High engagement:** particles settle into a more coherent, stable mesh.

Engagement does not resize the main particles or remove their minimum visibility. This preserves the independent heart-rate mapping.

This represents how strongly the audience member is held by the performance.

Physiological Standard Deviations

Tonic Electrodermal Activity Standard Deviation

This controls the slow, overall pressure of the mesh.

- **Higher positive values** compress the mesh inward and lift its centre upward.
- **Lower negative values** expand the mesh outward and let its centre sink or relax downward.

This represents slow bodily tension or release.

Temperature Rate-of-Change Standard Deviation

This controls particle hue, travelling colour waves, and directional trails.

- **Positive values** produce warm red/orange activity travelling from left to right, with trails offset above the particles.
- **Negative values** produce cool blue activity travelling from right to left, with trails offset below the particles.
- **Larger magnitude** makes the trails stronger and wider.
- **Strong changes** can trigger short bursts of travelling colour waves.

Temperature does not change the slow global form, the global particle-size baseline, or the particles' minimum opacity.

This represents thermal change becoming a visible wave or wash through the audience body.

Skin Conductance Response Frequency Standard Deviation

This controls sudden local events: rings, sparks, and short shocks in the mesh.

- **Higher magnitude** creates more frequent and stronger events.
- **Positive events** form expanding rings and push the local surface upward and outward.
- **Negative events** contract inward and pull the local surface downward.
- **Strong events** brighten nearby particles and can create white spark activity.

SCR events do not change the global size of the main particles.

This represents moments of reaction, surprise, stress, or heightened attention.

Heart-Rate Standard Deviation

This controls the global size and synchronized pulse of every main particle, together with an upper semicircular halo above each of the four corner particles.

- **Positive values** enlarge every particle, quicken the pulse, and place each active halo outside its neutral reference ring.
- **Negative values** reduce every particle, slow the pulse, and place each active halo inside its neutral reference ring.
- **Positive pulses** expand outward; **negative pulses** contract inward.
- The active halos never cross their neutral reference rings, so the sign remains unambiguous during a pulse.
- All main particles and all four active halos share one pulse clock and remain in unison.
- Each halo follows its rendered corner particle with slight centre smoothing, while the pulse itself remains synchronized.
- If heart-rate data becomes stale, particles return to neutral and the active halos fade while the reference rings remain.

This represents the visible tempo of audience physiology.

Streams Not Used by This Visual

Arousal, valence, and inter-beat interval may still appear in shared compatibility contracts or incoming payloads. They do not drive the current abstract particle mesh or objective five-graph view.